

ACCESS MODIFIERS

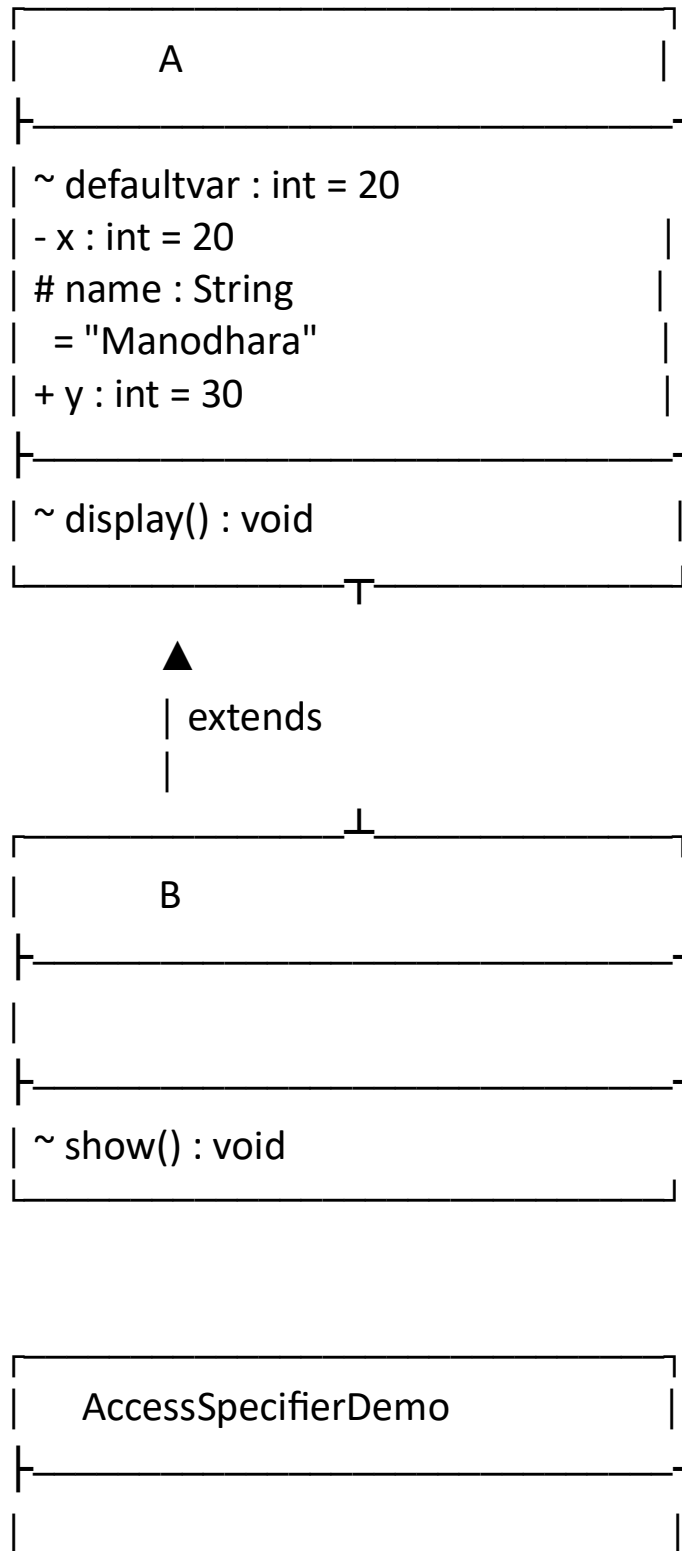
Objective :

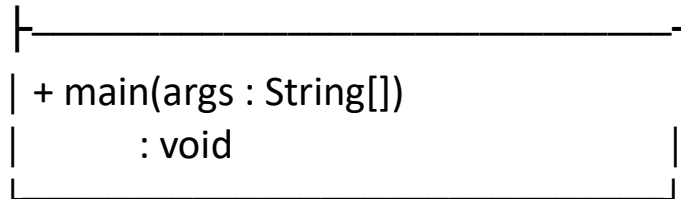
write the program using different kind of access modifiers
(public ,private,default,protected)

Algorithm :

1. Start.
2. Define a class A.
3. Declare a default variable defaultvar with value 20.
4. Declare a private variable x with value 20.
5. Declare a protected variable name with value "Manodhara".
6. Declare a public variable y with value 30.
7. Define the display() method in class A.
8. Display the heading "PARENT CLASS".
9. Display the values of defaultvar, x, name, and y.
10. Define a class B that extends class A.
11. Define the show() method in class B.
12. Display the heading "CHILD CLASS".
13. Access and display the default variable defaultvar.
14. Access and display the protected variable name.
15. Access and display the public variable y.
16. Do not access the private variable x directly in class B.
17. In the main() method, create an object obj of class B.
18. Call obj.show() to display the child class details.
19. Call obj.display() to display the parent class details.
20. Stop.

UML DIAGRAM :





SOURCE CODE :

```
package accessmodifiers;
```

```
class A{
    int defaultvar=20;
    private int x =20;
    protected String name = "Manodhara";
    public int y=30;
    void display(){
        System.out.println("-----PARENT CLASS-----");
        System.out.println("the value of defaultvar is :"+defaultvar);
        System.out.println("the value of x is :"+x);
        System.out.println("the name of a person is :"+name);
        System.out.println("the value of y is :"+y);
    }
}

class B extends A{
    void show(){
        System.out.println("-----CHILD CLASS-----");
        //default variable can accessed
        System.out.println("the value of defaultvar is :"+defaultvar);
        //private variable cannot be accessed to other child class
        // System.out.println("the value of x is :"+x);
        //protected variable also accessed
        System.out.println("the name of a person is :"+name);
        //public variable accessed anywhere
        System.out.println("the value of y is :"+y);
    }
}
```

```
public class AccessSpecifierDemo {
    public static void main(String[] args) {
```

```
        B obj =new B();  
        obj.show();  
        obj.display();  
    }  
}
```

OUTPUT :

```
-C:\Program Files\Java\jdk-24\bin\java.  
-----CHILD CLASS-----  
the value of defaultvar is :20  
the name of a person is :Manodhara  
the value of y is :30  
-----PARENT CLASS-----  
the value of defaultvar is :20  
the value of x is :20  
the name of a person is :Manodhara  
the value of y is :30  
  
Process finished with exit code 0
```